



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 08/07/2026

EXPERIMENT 1 -- DDL Commands (CREATE, ALTER, DROP, TRUNCATE, RENAME)

AIM:

To create and manage the database structure for a **Student CGPA** using Data Definition Language (DDL) commands such as **CREATE, ALTER, DROP, TRUNCATE, and RENAME**.

PROCEDURE:

1. Open MySQL and create the database for the Student CGPA.
2. Use the CREATE command to create required tables such as Student, Reg no., Department, number of subjects, number of pass subjects, number of fail subjects and CGPA with suitable attributes and data types.
3. Use the ALTER command to add, modify, or remove columns from an existing table.
4. Use the RENAME command to change the name of an existing table.
5. Use the TRUNCATE command to remove all records from a table while keeping its structure.
6. Use the DROP command to permanently delete an unwanted table from the database.
7. Use DESC table_name; or SHOW TABLES; after the operations to verify the changes.
8. Execute each command and observe the output to ensure that all DDL operations are performed successfully.

PROGRAM:

1. CREATE DATABASE

CREATE DATABASE Student_CGPA;

Then select the database:

USE Student_CGPA;

OUTPUT:

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
✓ 1	00:48:18	CREATE DATABASE Student_CGPA	1 row(s) affected	0.000 sec
✓ 2	00:48:18	USE Student_CGPA	0 row(s) affected	0.000 sec

2. CREATE TABLE

```
CREATE TABLE student_details (  
    student_id INT PRIMARY KEY,  
    student_name VARCHAR(50),  
    department VARCHAR(50),  
    semester INT,  
    cgpa DECIMAL(3,2)  
);
```

OUTPUT:

The screenshot shows the 'Output' window of a database management tool. It displays three actions executed successfully:

#	Time	Action	Message	Duration / Fetch
1	00:48:18	CREATE DATABASE Student_CGPA	1 row(s) affected	0.000 sec
2	00:48:18	USE Student_CGPA	0 row(s) affected	0.000 sec
3	00:49:57	CREATE TABLE Student (Student_ID INT PRIMARY KEY, Student_Name VARCHAR(50), Department V...	0 row(s) affected	0.031 sec

Below the output window, a SQL query is entered in the editor:

```
10 (104, Priya, 21, 9.25);  
17 • SELECT * FROM student_details;
```

The 'Result Grid' shows the data returned by the query:

student_id	student_name	department	semester	cgpa
101	Deepthi	CSE	5	8.75
102	Anu	CSE	5	9.10
103	Rahul	ECE	5	8.50
104	Priya	IT	5	9.25
NULL	NULL	NULL	NULL	NULL

3. ALTER – ADD COLUMN

Add a new column called E-mail:

```
ALTER TABLE student_details  
ADD email VARCHAR(50);  
DESC student_details;
```

OUTPUT:

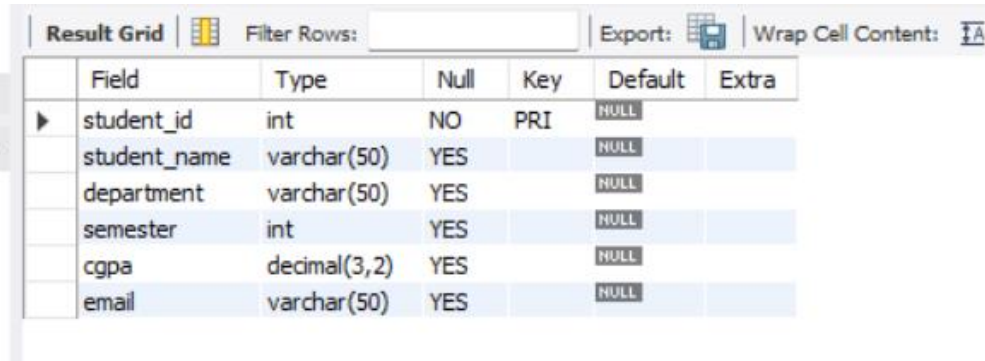
The screenshot shows the 'Result Grid' displaying the structure of the 'student_details' table after the ALTER command:

Field	Type	Null	Key	Default	Extra
student_id	int	NO	PRI	NULL	
student_name	varchar(50)	YES		NULL	
department	varchar(50)	YES		NULL	
semester	int	YES		NULL	
cgpa	decimal(3,2)	YES		NULL	
email	varchar(50)	YES		NULL	

4. ALTER – MODIFY COLUMN

Change the size of Student_Name:

```
ALTER TABLE student_details  
MODIFY student_name VARCHAR(100);  
DESC student_details;
```



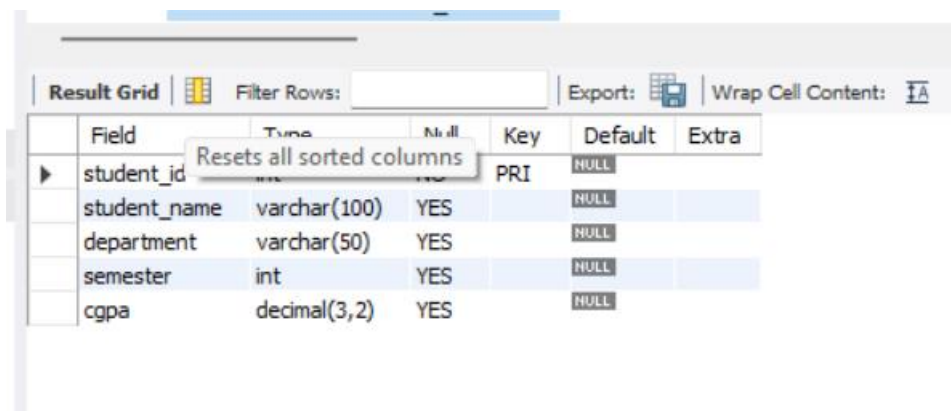
	Field	Type	Null	Key	Default	Extra
▶	student_id	int	NO	PRI	NULL	
	student_name	varchar(50)	YES		NULL	
	department	varchar(50)	YES		NULL	
	semester	int	YES		NULL	
	cgpa	decimal(3,2)	YES		NULL	
	email	varchar(50)	YES		NULL	

5. ALTER – DROP COLUMN

Remove the E- MAIL column:

```
ALTER TABLE student_details  
DROP COLUMN email;  
DESC student_details;
```

OUTPUT:



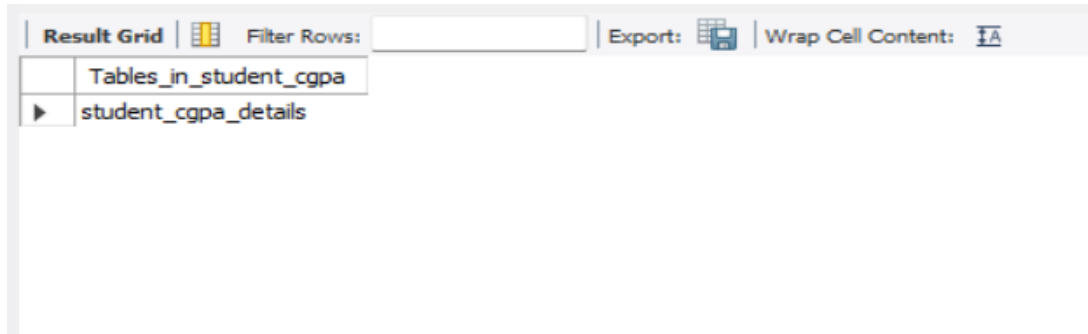
	Field	Type	Null	Key	Default	Extra
▶	student_id	int	NO	PRI	NULL	
	student_name	varchar(100)	YES		NULL	
	department	varchar(50)	YES		NULL	
	semester	int	YES		NULL	
	cgpa	decimal(3,2)	YES		NULL	

6. RENAME TABLE

Rename Flight to student_cgpa_details:

```
ALTER TABLE student_details  
RENAME TO student_cgpa_details;  
SHOW TABLES;
```

OUTPUT:



	Tables_in_student_cgpa
▶	student_cgpa_details

6. TRUNCATE TABLE

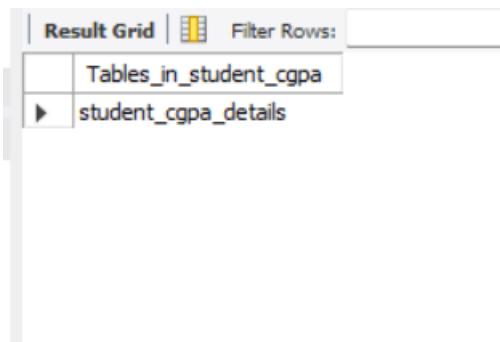
```
TRUNCATE TABLE student_cgpa_details;
```

Check:

```
SELECT * FROM student_cgpa_details;
```

The table will be empty, but the table still exists

OUTPUT:



	Tables_in_student_cgpa
▶	student_cgpa_details

7. DROP TABLE:

```
DROP TABLE student_cgpa_details;  
SHOW TABLES;
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

Tables_in_student_cgpa
student_cgpa_details

RESULT: Thus the DDL commands executed successfully.